

Integrating Cognitive Behavioral Therapy and Heart Rate Variability Biofeedback in Virtual Reality, Augmented Reality, and Mixed Reality as a Mental Health Intervention

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ABSTRACT

We introduce an innovative approach to reducing cognitive stress by integrating two non-pharmacological therapeutic strategies, Cognitive Behavioral Therapy (CBT) videos and Heart Rate Variability Biofeedback (HRV-BF), within Virtual Reality (VR), Augmented Reality (AR) and Mixed Reality (MR). The research demo presents a virtually created mental health therapist situated within a virtual therapy office who will deliver evidence-based CBT, which has been found to be efficacious in reducing stress. The simulated evidence-based HRV-BF strategy also aims to reduce stress by engaging the Parasympathetic Nervous System. The aim is to demonstrate the potentiality of advanced immersive technologies to deliver engaging and efficacious stress management interventions. The findings from this research will be pivotal in evaluating the comparative effectiveness of CBT and HRV-BF across the VR, AR, and MR environments.

Keywords: Virtual Reality, Augmented Reality, Mixed Reality, Spatial Computing, Digital Mental Health

1 INTRODUCTION

Cognitive stress can lead to a spectrum of physiological and psychological disorders when it remains unmitigated [1]. Chronic exposure is known to activate the Sympathetic Nervous System (SNS), resulting in a variety of pathological manifestations such as heart arrhythmias, anxiety, bruxism, cephalgia, insomnia, dysthymia, hypertension, and cardiovascular pathologies [2]. Conventional stress reduction interventions include mental health therapy, self-care, relaxation techniques (diaphragmatic breathing and progressive muscle relaxation), and mindful exercises [2, 3]. Technological advancements provide diverse methods for stress reduction, allowing for immediate and low-maintenance aid [4]. For example, the unique immersive capability of virtual reality (VR) has been demonstrated to complement or potentially serve as a substitute for conventional therapy and stress management [3, 4]. Few researchers have explored the utilization of alternative immersive environments, such as Augmented Reality (AR) and Mixed Reality (MR), to showcase their potential for stress reduction and to show a comparative analysis of the effectiveness among the distinctive immersive technologies. To address

this research gap, we developed a stress management intervention through immersive applications of Cognitive Behavioral Therapy (CBT) [5] and Heart Rate Variability Biofeedback (HRV-BF) [6]. CBT is known for its ability to provide cognitive restructuring and behavioral strategies [5]. Concurrently, HRV-BF enables real-time physiological feedback for effective stress symptom regulation [7]. Research shows that 2D computer-based stress-management interventions can reduce stress over six months [8]. In contrast to these conventional computer-based interventions, this research explores the unique benefits that VR, AR, and MR bring to the forefront in stress management.

2 INTERVENTION DEVELOPMENT

The HRV-BF intervention incorporates an immersive virtual beach experience, allowing participants to virtually engage with an interactive, natural environment [7]. Research has shown that an immersive natural environment, such as a beach, positively affects stress reduction based on attention restoration theory and affective computing techniques [7]. The VR intervention utilized the Meta Quest 2, the AR intervention utilized the Vuzix Blade, and the MR used the Microsoft HoloLens 2, allowing users to interact with virtual hologram-styled projections. These interventions aim to create engaging and therapeutic content for stress reduction.

2.1 Beach, Therapy Room, and HRV Biofeedback

The virtual beach scene, therapy office, and Biofeedback systems were integrated in Unity 2022.3.2f utilizing the 3D Built-In Rendering Systems template (see Figure 1 and Figure 2). A 3D-rendered

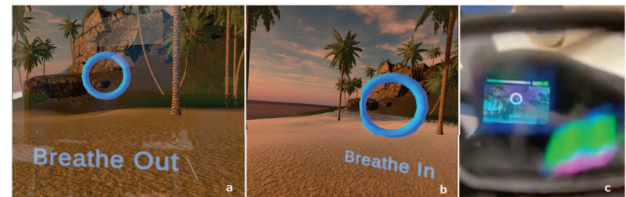


Figure 1: Beach Scene in a) MR, b) VR, and c) AR

biofeedback breathing circle simulation was created to coach users in real-time, helping them achieve an effective breathing rate and optimally engage their nervous system. The animation scales the X, Y, and Z axes (see Figure 1). A breathing pattern of 5.5 breaths per minute with an Inhalation-to-Exhalation ratio of 5:5 resulted in a higher HRV [7]. This breathing rate has been shown to engage the Parasympathetic Nervous System (PNS) by increasing HRV and inducing relaxation [7, 9]. In designing a virtual therapist office, we sought to enhance user interaction by providing an immersive environment for receiving CBT. A pre-built room asset (via the Unity Asset Store) and living room furniture asset (via WaveFunctionVR) were used to create the virtual therapist office. In Figure 2, the

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Microsoft HoloLens 2 (MR) therapist delivers CBT in the user's physical space, perceiving the therapist in the same room.



Figure 2: Holographic Therapists placed in a) MR device Microsoft HoloLens 2, and b) VR device Meta Quest 2.

2.2 Integration of Volumetric Video

Volumetric video records 3D spaces utilizing one or multiple depth camera sensors to capture true-to-life colors, textures, and depth information for spatialized immersion [10]. The result is a 3D model, enabling viewers to explore and interact with the virtual scene from various perspectives. The 3D virtual therapist was generated in this demo using Depthkit Core's volumetric video in conjunction with a Microsoft Azure Kinect camera (as shown in Figure 3). The Depthkit Expansion Package for Unity was initially used to integrate the video capture on the Meta Quest 2 device but faced compatibility issues with the Microsoft HoloLens 2. Importing the OBJ file sequence into Unity and playing it as an animation by displaying each OBJ file as a video frame proved effective for integrating Depthkit's volumetric video into Microsoft HoloLens 2. Nevertheless, this approach had drawbacks, as the sequence of OBJ files required much more storage space than the original plugin's combined-per-pixel video, which includes depth and color data in a single video file optimized for Unity. This was rectified by accessing a previous version of the Depthkit Expansion Package for Unity that worked with the Microsoft HoloLens 2. The previous plugin addressed compatibility concerns and offered higher quality than the OBJ file sequence while consuming less storage space.

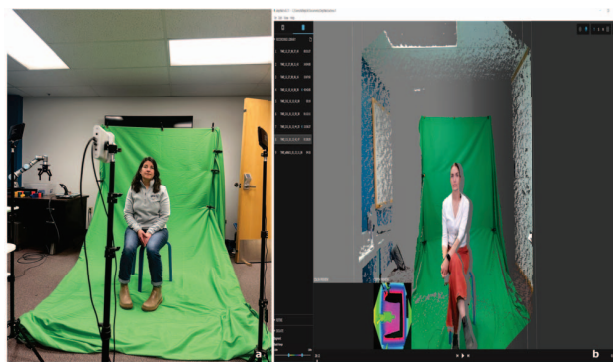


Figure 3: a) Green screen with lighting for capturing a model with Depthkit and one Microsoft Azure Kinect depth camera b) a model in the Depthkit's interface while recording volumetric video.

Future work includes integrating the interventions into VR, AR, and MR to collect objective data (e.g., HRV, electrodermal activity, and respiration rate) and subjective data using validated survey instruments to measure perceived stress, emotion, presence, engagement, and usability. Additionally, an efficacy evaluation will be conducted

in collaboration with Prisma Health's Addiction Center in Greenville, SC.

3 CONCLUSION

Integrating novel technologies, including VR, AR, and MR, with therapeutic strategies such as CBT videos and HRV-BF techniques constructs an advancement in stress management interventions and a means to investigate the potential for non-pharmacological interventions. With increased user engagement in immersive technology, we aim to enhance users' unique ability of interactivity in a meaningful way to address stress-related challenges. This technological integration demonstrates the potential usage of virtual therapist-guided CBT and HRV-BF for stress reduction. Moreover, it also opens avenues for further research and development in immersive interventions for improving mental health.

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